

**Table 1** Schedule of events for immune profiling studies

| Day:                                         | -28 to -7 | 0 | 1 | 3 | 7 | 14 | 28 | 29 | 32 | 35 | 42 | 56 |
|----------------------------------------------|-----------|---|---|---|---|----|----|----|----|----|----|----|
| <b>Study injection</b> (Heplisav or Engerix) |           | X |   |   |   |    | X  |    |    |    |    |    |
| <b>Serum Assays: 1x8ml vacutainer</b>        |           |   |   |   |   |    |    |    |    |    |    |    |
| Serology (anti-HBsAg)                        |           | X |   |   |   | X  | X  |    |    |    | X  | X  |
| Serum cytokines                              | X         | X | X | X | X | X  | X  | X  | X  | X  | X  | X  |
| <b>RNA Assays: 1x3 ml Tempus tubes</b>       |           |   |   |   |   |    |    |    |    |    |    |    |
| Whole blood microarray                       | X         | X | X | X | X | X  | X  | X  | X  | X  | X  | X  |
| <b>PBMC Assays: 3x8 ml CPT tubes</b>         |           |   |   |   |   |    |    |    |    |    |    |    |
| PBMC for phenotyping                         | X         | X |   |   | X | X  | X  |    |    | X  | X  | X  |
| PBMC for Restimulation                       |           |   |   |   |   |    |    |    |    |    |    |    |
| FACS with CD154 gating                       |           | X |   |   | X | X  | X  |    |    | X  | X  | X  |
| PBMC for Restimulation                       |           |   |   |   |   |    |    |    |    |    |    |    |
| Gene expression by Nanostring                |           | X |   |   | X | X  | X  |    |    | X  | X  | X  |
| PBMC Stimulation with                        |           |   |   |   |   |    |    |    |    |    |    |    |
| innate immune ligands                        | X         |   |   |   |   |    |    |    |    |    |    |    |

**Assays overview**

| Cohort                        | COHORT 1+2    |    |    |    |               |      |      |      |       |               |     |
|-------------------------------|---------------|----|----|----|---------------|------|------|------|-------|---------------|-----|
| Visit                         | V1            | V2 | V3 | V4 | V5            | V6   | V7   | V8   | V9    | V10           | V11 |
| Day                           | D0            | 1  | 3  | 7  | 28/0          | 29/1 | 31/3 | 35/7 | 56/28 | 156           | 196 |
| Vaccine: HepB-alumn (Engerix) | 1st injection |    |    |    | 2nd injection |      |      |      |       | 3rd injection |     |

|                           |                                                                                          |                     |                     |                     |                     |                     |                     |                     |                     |            |            |
|---------------------------|------------------------------------------------------------------------------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|------------|------------|
| Sample collection         | serum, tempus, PBMC                                                                      | serum, tempus, PBMC | serum, tempus, PBMC | serum, tempus, PBMC | serum, tempus, PBMC | serum, tempus, PBMC | serum, tempus, PBMC | serum, tempus, PBMC | serum, tempus, PBMC | serum only | serum only |
| Assay                     | Sample ID                                                                                |                     |                     |                     |                     |                     |                     |                     |                     |            |            |
| Clinical data             | demographics                                                                             |                     |                     |                     |                     |                     |                     |                     |                     |            |            |
| CBC                       | n.a.                                                                                     | n.a.                | n.a.                | n.a.                | n.a.                | n.a.                | n.a.                | n.a.                | n.a.                | n.a.       | n.a.       |
| Microarray (WB, illumina) | cohort 1+2 all samples, all time points hybridized, in analysis                          |                     |                     |                     |                     |                     |                     |                     |                     |            |            |
| Flow cytometry            | cyropreserved PBMC; for all time points in a subset of 12 donors; data not yet generated |                     |                     |                     |                     |                     |                     |                     |                     |            |            |
| Serology                  | binding antibody titers for D0 and Dxxx available                                        |                     |                     |                     |                     |                     |                     |                     |                     |            |            |
| other                     | n.a.                                                                                     |                     |                     |                     |                     |                     |                     |                     |                     |            |            |
| in vitro                  | CD154 assay on cryopreserved PBMC                                                        |                     |                     |                     |                     |                     |                     |                     |                     |            |            |

### Study Design

|                               |            | COHORT 1+2          |                     |                     |                     |                     |                     |                     |                     |                     |               |            |
|-------------------------------|------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------|------------|
|                               | Visit      | V1                  | V2                  | V3                  | V4                  | V5                  | V6                  | V7                  | V8                  | V9                  | V10           | V11        |
|                               | Day        | D0                  | 1                   | 3                   | 7                   | 28/0                | 29/1                | 31/3                | 35/7                | 56/28               | 156           | 196        |
| Vaccine: HepB-alumn (Engerix) |            | 1st injection       |                     |                     |                     | 2nd injection       |                     |                     |                     |                     | 3rd injection |            |
|                               | Sample co  | serum, tempus, PBMC | serum, tempus, PBMC | serum, tempus, PBMC | serum, tempus, PBMC | serum, tempus, PBMC | serum, tempus, PBMC | serum, tempus, PBMC | serum, tempus, PBMC | serum, tempus, PBMC | serum only    | serum only |
| Cohort                        | Patient ID | Sample ID           | Sample ID           | Sample ID           | Sample ID           | Sample ID           | Sample ID           | Sample ID           | Sample ID           | Sample ID           | Sample ID     | Sample ID  |
| Cohort 1                      | 001        | 001-V1              | 001-V2              | 001-V3              | 001-V4              | 001-V5              | 001-V6              | 001-V7              | 001-V8              | 001-V9              | 001-V10       | 001-V11    |
| Cohort 1                      | 002        | 002-V1              | 002-V2              | 002-V3              | 002-V4              | 002-V5              | 002-V6              | 002-V7              | 002-V8              | 002-V9              | 002-V10       | 002-V11    |
| Cohort 1                      | 003        | 003-V1              | 003-V2              | 003-V3              | 003-V4              | 003-V5              | 003-V6              | 003-V7              | 003-V8              | 003-V9              | 003-V10       | 003-V11    |
| Cohort 1                      | 004        | 004-V1              | 004-V2              | 004-V3              | 004-V4              | 004-V5              | 004-V6              | 004-V7              | 004-V8              | 004-V9              | 004-V10       | 004-V11    |
| Cohort 1                      | 005        | 005-V1              | 005-V2              | 005-V3              | 005-V4              | 005-V5              | 005-V6              | 005-V7              | 005-V8              | 005-V9              | 005-V10       | 005-V11    |
| Cohort 1                      | 007        | 007-V1              | 007-V2              | 007-V3              | 007-V4              | 007-V5              | 007-V6              | 007-V7              | 007-V8              | 007-V9              | 007-V10       | 007-V11    |
| Cohort 1                      | 008        | 008-V1              | 008-V2              | 008-V3              | 008-V4              | 008-V5              | 008-V6              | 008-V7              | 008-V8              | 008-V9              | 008-V10       | 008-V11    |
| Cohort 1                      | 009        | 009-V1              | 009-V2              | 009-V3              | 009-V4              | 009-V5              | 009-V6              | 009-V7              | 009-V8              | 009-V9              | 009-V10       | 009-V11    |
| Cohort 1                      | 010        | 010-V1              | 010-V2              | 010-V3              | 010-V4              | 010-V5              | 010-V6              | 010-V7              | 010-V8              | 010-V9              | 010-V10       | 010-V11    |
| Cohort 1                      | 011        | 011-V1              | 011-V2              | 011-V3              | 011-V4              | 011-V5              | 011-V6              | 011-V7              | 011-V8              | 011-V9              | 011-V10       | 011-V11    |
| Cohort 1                      | 012        | 012-V1              | 012-V2              | 012-V3              | 012-V4              | 012-V5              | 012-V6              | 012-V7              | 012-V8              | 012-V9              | 012-V10       | 012-V11    |
| Cohort 1                      | 013        | 013-V1              | 013-V2              | 013-V3              | 013-V4              | 013-V5              | 013-V6              | 013-V7              | 013-V8              | 013-V9              | 013-V10       | 013-V11    |
| Cohort 1                      | 014        | 014-V1              | 014-V2              | 014-V3              | 014-V4              | 014-V5              | 014-V6              | 014-V7              | 014-V8              | 014-V9              | 014-V10       | 014-V11    |
| Cohort 1                      | 015        | 015-V1              | 015-V2              | 015-V3              | 015-V4              | 015-V5              | 015-V6              | 015-V7              | 015-V8              | 015-V9              | 015-V10       | 015-V11    |
| Cohort 1                      | 016        | 016-V1              | 016-V2              | 016-V3              | 016-V4              | 016-V5              | 016-V6              | 016-V7              | 016-V8              | 016-V9              | 016-V10       | 016-V11    |
| Cohort 1                      | 017        | 017-V1              | 017-V2              | 017-V3              | 017-V4              | 017-V5              | 017-V6              | 017-V7              | 017-V8              | 017-V9              | 017-V10       | 017-V11    |
| Cohort 1                      | 019        | 019-V1              | 019-V2              | 019-V3              | 019-V4              | 019-V5              | 019-V6              | 019-V7              | 019-V8              | 019-V9              | 019-V10       | 019-V11    |
| Cohort 1                      | 020        | 020-V1              | 020-V2              | 020-V3              | 020-V4              | 020-V5              | 020-V6              | 020-V7              | 020-V8              | 020-V9              | 020-V10       | 020-V11    |
| Cohort 1                      | 022        | 022-V1              | 022-V2              | 022-V3              | 022-V4              | 022-V5              | 022-V6              | 022-V7              | 022-V8              | 022-V9              | 022-V10       | 022-V11    |
| Cohort 1                      | 023        | 023-V1              | 023-V2              | 023-V3              | 023-V4              | 023-V5              | 023-V6              | 023-V7              | 023-V8              | 023-V9              | 023-V10       | 023-V11    |
| Cohort 1                      | 024        | 024-V1              | 024-V2              | 024-V3              | 024-V4              | 024-V5              | 024-V6              | 024-V7              | 024-V8              | 024-V9              | 024-V10       | 024-V11    |

|          |     |        |        |        |        |        |        |        |        |        |         |         |
|----------|-----|--------|--------|--------|--------|--------|--------|--------|--------|--------|---------|---------|
| Cohort 1 | 029 | 029-V1 | 029-V2 | 029-V3 | 029-V4 | 029-V5 | 029-V6 | 029-V7 | 029-V8 | 029-V9 | 029-V10 | 029-V11 |
| Cohort 1 | 030 | 030-V1 | 030-V2 | 030-V3 | 030-V4 | 030-V5 | 030-V6 | 030-V7 | 030-V8 | 030-V9 | 030-V10 | 030-V11 |
| Cohort 1 | 039 | 039-V1 | 039-V2 | 039-V3 | 039-V4 | 039-V5 | 039-V6 | 039-V7 | 039-V8 | 039-V9 | 039-V10 | 039-V11 |
| Cohort 1 | 044 | 044-V1 | 044-V2 | 044-V3 | 044-V4 | 044-V5 | 044-V6 | 044-V7 | 044-V8 | 044-V9 | 044-V10 | 044-V11 |
|          |     |        |        |        |        |        |        |        |        |        |         |         |
| Cohort 2 | 026 | 026-V1 | 026-V2 | 026-V3 | 026-V4 | 026-V5 | 026-V6 | 026-V7 | 026-V8 | 026-V9 | 026-V10 | 026-V11 |
| Cohort 2 | 027 | 027-V1 | 027-V2 | 027-V3 | 027-V4 | 027-V5 | 027-V6 | 027-V7 | 027-V8 | 027-V9 | 027-V10 | 027-V11 |
| Cohort 2 | 028 | 028-V1 | 028-V2 | 028-V3 | 028-V4 | 028-V5 | 028-V6 | 028-V7 | 028-V8 | 028-V9 | 028-V10 | 028-V11 |
| Cohort 2 | 031 | 031-V1 | 031-V2 | 031-V3 | 031-V4 | 031-V5 | 031-V6 | 031-V7 | 031-V8 | 031-V9 | 031-V10 | 031-V11 |
| Cohort 2 | 032 | 032-V1 | 032-V2 | 032-V3 | 032-V4 | 032-V5 | 032-V6 | 032-V7 | 032-V8 | 032-V9 | 032-V10 | 032-V11 |
| Cohort 2 | 033 | 033-V1 | 033-V2 | 033-V3 | 033-V4 | 033-V5 | 033-V6 | 033-V7 | 033-V8 | 033-V9 | 033-V10 | 033-V11 |
| Cohort 2 | 034 | 034-V1 | 034-V2 | 034-V3 | 034-V4 | 034-V5 | 034-V6 | 034-V7 | 034-V8 | 034-V9 | 034-V10 | 034-V11 |
| Cohort 2 | 035 | 035-V1 | 035-V2 | 035-V3 | 035-V4 | 035-V5 | 035-V6 | 035-V7 | 035-V8 | 035-V9 | 035-V10 | 035-V11 |
| Cohort 2 | 036 | 036-V1 | 036-V2 | 036-V3 | 036-V4 | 036-V5 | 036-V6 | 036-V7 | 036-V8 | 036-V9 | 036-V10 | 036-V11 |
| Cohort 2 | 038 | 038-V1 | 038-V2 | 038-V3 | 038-V4 | 038-V5 | 038-V6 | 038-V7 | 038-V8 | 038-V9 | 038-V10 | 038-V11 |
| Cohort 2 | 040 | 040-V1 | 040-V2 | 040-V3 | 040-V4 | 040-V5 | 040-V6 | 040-V7 | 040-V8 | 040-V9 | 040-V10 | 040-V11 |
| Cohort 2 | 041 | 041-V1 | 041-V2 | 041-V3 | 041-V4 | 041-V5 | 041-V6 | 041-V7 | 041-V8 | 041-V9 | 041-V10 | 041-V11 |
| Cohort 2 | 042 | 042-V1 | 042-V2 | 042-V3 | 042-V4 | 042-V5 | 042-V6 | 042-V7 | 042-V8 | 042-V9 | 042-V10 | 042-V11 |
| Cohort 2 | 043 | 043-V1 | 043-V2 | 043-V3 | 043-V4 | 043-V5 | 043-V6 | 043-V7 | 043-V8 | 043-V9 | 043-V10 | 043-V11 |
| Cohort 2 | 046 | 046-V1 | 046-V2 | 046-V3 | 046-V4 | 046-V5 | 046-V6 | 046-V7 | 046-V8 | 046-V9 | 046-V10 | 046-V11 |
| Cohort 2 | 047 | 047-V1 | 047-V2 | 047-V3 | 047-V4 | 047-V5 | 047-V6 | 047-V7 | 047-V8 | 047-V9 | 047-V10 | 047-V11 |
| Cohort 2 | 048 | 048-V1 | 048-V2 | 048-V3 | 048-V4 | 048-V5 | 048-V6 | 048-V7 | 048-V8 | 048-V9 | 048-V10 | 048-V11 |
| Cohort 2 | 049 | 049-V1 | 049-V2 | 049-V3 | 049-V4 | 049-V5 | 049-V6 | 049-V7 | 049-V8 | 049-V9 | 049-V10 | 049-V11 |
| Cohort 2 | 050 | 050-V1 | 050-V2 | 050-V3 | 050-V4 | 050-V5 | 050-V6 | 050-V7 | 050-V8 | 050-V9 | 050-V10 | 050-V11 |
| Cohort 2 | 051 | 051-V1 | 051-V2 | 051-V3 | 051-V4 | 051-V5 | 051-V6 | 051-V7 | 051-V8 | 051-V9 | 051-V10 | 051-V11 |
| Cohort 2 | 052 | 052-V1 | 052-V2 | 052-V3 | 052-V4 | 052-V5 | 052-V6 | 052-V7 | 052-V8 | 052-V9 | 052-V10 | 052-V11 |
| Cohort 2 | 054 | 054-V1 | 054-V2 | 054-V3 | 054-V4 | 054-V5 | 054-V6 | 054-V7 | 054-V8 | 054-V9 | 054-V10 | 054-V11 |
| Cohort 2 | 055 | 055-V1 | 055-V2 | 055-V3 | 055-V4 | 055-V5 | 055-V6 | 055-V7 | 055-V8 | 055-V9 | 055-V10 | 055-V11 |
| Cohort 2 | 058 | 058-V1 | 058-V2 | 058-V3 | 058-V4 | 058-V5 | 058-V6 | 058-V7 | 058-V8 | 058-V9 | 058-V10 | 058-V11 |
| Cohort 2 | 060 | 060-V1 | 060-V2 | 060-V3 | 060-V4 | 060-V5 | 060-V6 | 060-V7 | 060-V8 | 060-V9 | 060-V10 | 060-V11 |